Medical Imaging Roadmap

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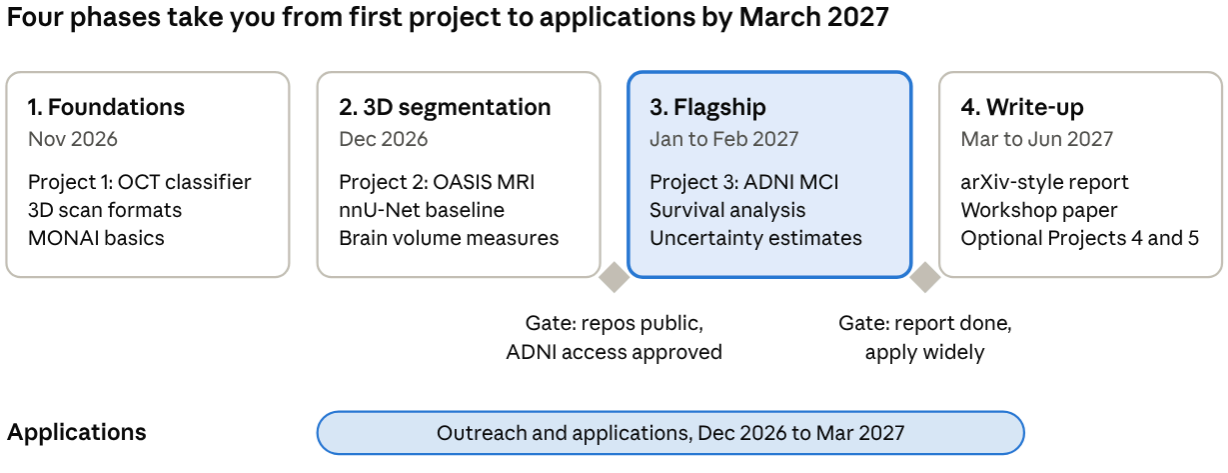
Goal and timeline

By March 2027, have three medical imaging projects on GitHub and applications out to funded medical imaging PhDs, for the Fall 2027 intake. You already know deep learning and computer vision. What's missing is the medical side: 3D scan formats, preprocessing, patient-level evaluation, survival analysis and data-access paperwork.

- **Time budget:** about 12–15 hours a week alongside your job, for roughly 5 months of focused work (November 2026 to March 2027).
- **One research story across both fields:** making model confidence useful to people. Your caption project shows ASR confidence to DHH viewers. In medical imaging, the same idea becomes calibrated, uncertainty-aware predictions a clinician can trust. Every project below includes a calibration step so the story holds together.
- **Keep the HCI track alive:** this is a second track for funded positions, not a replacement. Keep emailing captioning and accessibility labs in parallel.

Roadmap

Four phases build the skills in order, and outreach starts once you have two public repos to show.



roadmap · 4 phases, 2 gates, Nov 2026 to Jun 2027

The flagship phase is highlighted because it produces the work that gets you interviews. It needs ADNI access approved before January, so send the request in week 1.

What to learn, and where

Learn each skill by using it in the project it belongs to, not by finishing courses first. Everything below is free.

Skill	Why it matters	Where to learn it	Used in
3D scan files (NIfTI, DICOM), voxel spacing, orientation	Scans are 3D volumes with physical units, not photos	NiBabel docs, SimpleITK notebooks, 3D Slicer for viewing	Projects 2–5
MONAI (PyTorch for medical imaging)	The standard library; reviewers and labs expect it	MONAI tutorials, start with the 2D classification and 3D segmentation notebooks	All projects
nnU-Net	The segmentation baseline every paper compares against	nnU-Net GitHub and its documentation	Projects 2 and 4
Preprocessing: skull stripping, intensity correction, registration, CT windowing	Most results depend on getting this right	FSL course (brain MRI), ANTsPy	Projects 2, 3 and 5
Patient-level splits, Dice, AUC with confidence intervals, calibration	Leakage from image-level splits is the most common mistake newcomers make	Metrics Reloaded (which metric to use when)	All projects
Survival analysis: Cox models, C-index, censoring	Needed to predict whether and when a disease progresses, which is the core of Popuri-style outcome prediction	lifelines docs, scikit-survival	Project 3
Uncertainty: MC dropout, deep ensembles, temperature scaling	Your research thread across captions and imaging	Your existing ML knowledge plus the MONAI and PyTorch docs	Projects 1 and 3
Basic anatomy and clinical terms	Needed to read papers and talk to clinicians	Radiopaedia for brain, retina and abdominal CT	Throughout

One structured course alongside it (optional): Stanford's AI in Healthcare lectures or a Coursera medical imaging course, at one lecture a week. Don't let a course replace building the projects.

Portfolio projects

Three core projects are enough to apply. Each gets its own GitHub repo with a README, results table, figures and a short write-up of what failed. Projects 4 and 5 are optional extras.

#	Project	Data (where to get it)	Tools	Deliverable	Time
1	**Retinal OCT classification with calibrated confidence**: classify CNV, DME, drusen and normal; show that low-confidence cases really are the hard ones	Kermany OCT dataset (free) or OCTMNIST for a first pass	PyTorch, MONAI, Grad-CAM, temperature scaling	Repo + reliability diagram + Grad-CAM figures	2 weeks
2	**3D brain MRI segmentation**: segment tissue or the hippocampus and measure volumes	OASIS (free, simple sign-up)	NiBabel, ANTsPy, MONAI, nnU-Net	Repo + nnU-Net comparison + volume plots	3 weeks
3	**Flagship: predicting conversion from mild cognitive impairment to Alzheimer's**, with uncertainty	ADNI (apply for access; approval takes time)	MONAI 3D CNN, lifelines or scikit-survival, MC dropout or ensembles	Repo + 4–6 page arXiv-style report; the piece to show Popuri-type labs	6–8 weeks
4	Optional: **body CT organ segmentation and body-composition measures**	TotalSegmentator dataset, Medical Segmentation Decathlon	nnU-Net, TotalSegmentator	Repo + muscle and fat measures per patient	2–3 weeks
5	Optional: **tracking change over time**, using repeat scans of the same patients (atrophy rates, matching scans to the same person)	ADNI or OASIS-2 longitudinal scans	Registration (ANTsPy), Siamese networks from your re-ID work	Repo + short write-up linking it to your wildlife re-ID experience	3–4 weeks

Write-up target: turn Project 3 into a workshop paper or short paper. Check the current year's dates for MIDL short papers and MICCAI workshops (for example UNSURE, on uncertainty) on their official sites, since deadlines move each year.

Where to apply

The best fits are groups that do neuroimaging or Alzheimer's prediction, because Project 3 speaks directly to them. In the Netherlands and Germany, PhD positions are usually paid jobs with no tuition, which fits your funding constraint. Positions open year-round, so watch the vacancy portals rather than waiting for one deadline.

Group	Country	Why it fits you	Where positions appear
Karteek Popuri, Memorial University	Canada (you'd ruled it out)	Advertising fully funded PhDs in outcome prediction from brain scans, CT and retinal OCT, which match Projects 1–4	MUN recruitment page; email him directly
Biomedical Imaging Group Rotterdam, Erasmus MC	Netherlands	Large neuroimaging and dementia imaging group	AcademicTransfer, Erasmus MC careers
Diagnostic Image Analysis Group, Radboudumc	Netherlands	One of the largest medical imaging AI groups (retina, CT, pathology)	AcademicTransfer, DIAG vacancies page
Image Sciences Institute, UMC Utrecht	Netherlands	Brain MRI and CT image analysis	AcademicTransfer
Division of Medical Image Computing, DKFZ Heidelberg	Germany	The nnU-Net group; segmentation and uncertainty	DKFZ jobs portal, Helmholtz Information and Data Science School for Health
Lab for AI in Medicine, TU Munich	Germany	Medical imaging deep learning and privacy-preserving AI	Lab site, EURAXESS

Group	Country	Why it fits you	Where positions appear
ARAMIS team, Inria / Paris Brain Institute	France	Alzheimer's prediction from brain imaging; builds the open-source Clinica pipeline	Inria jobs, EURAXESS
Epione team, Inria Sophia Antipolis	France	Computational medical imaging and modelling	Inria jobs
Center for Machine Vision and Signal Analysis, University of Oulu	Finland	Computer vision with medical imaging work; open calls via the doctoral school	University of Oulu careers, EURAXESS
KAIST Bio and Brain Engineering / AI groups	South Korea	Strong medical imaging AI	KAIST admissions; Global Korea Scholarship for funding

Open questions: MUN's graduate deadline, stipend and whether you need a supervisor's agreement before applying. Its website was down on 22 Oct 2026, so check again before emailing Popuri.

Application checklist

Start the slow items this week: data access and references take the longest.

This week

- [ ] Apply for ADNI data access (Project 3 depends on it)
- [ ] Register for OASIS and download the Kermany OCT dataset
- [ ] Set up a GitHub repo template: README, environment file, results table, figures folder

Before emailing any medical imaging professor (after Projects 1 and 2)

- [ ] Rewrite your CV with a "Medical imaging" projects section above the HCI projects for these applications
- [ ] Write a one-page research statement around "making model confidence useful to people": captions, then imaging
- [ ] Ask your references, Dr. Shatabda and Dr. Ahmmed, whether they can mention your ML and computer vision work

Outreach (from December 2026)

- [ ] Email Popuri once Project 2 is done, attaching the repos and naming his OCT and neuroimaging work
- [ ] Set saved searches for "medical image" PhD positions on AcademicTransfer, EURAXESS and Inria jobs
- [ ] Email two or three of the groups above, each with one sentence tied to a specific paper of theirs
- [ ] Apply to every open position that fits within a week of it appearing; many close in 4–6 weeks

Paperwork

- [ ] Check English test rules per position. Your IELTS Writing band of 6.0 is the risk point where a 6.5 minimum per band applies
- [ ] Look up Global Korea Scholarship and DAAD timelines if you apply in Korea or Germany